# Technical Documentation

ACDT Project 1, GROUP32: Is daily social media use associated with student mental health?
**Interactive data story (scroll, filters, calculators):** dhkim060225-web.github.io/group32-sns-mental-health
Code: github.com/dhkim060225-web/group32-sns-mental-health

## 1. Research question and hypotheses

Outcome Y = Mental_Health_Score; treatment X = Avg_Daily_Usage_Hours. Model: $Y = \beta_0 + \beta_1 X + \varepsilon$. $H_0$: $\beta_1 = 0$; $H_1$: $\beta_1 \neq 0$. Our earlier version was one-sided ($\beta_1 < 0$, controlling for academic level). We switched to two-sided because the regression framework tests $\beta_1 \neq 0$, a two-sided test is the more cautious choice, and with reverse causation possible we did not want to rule out either direction in advance. **$\alpha = 0.05$**: a Type I error would justify blanket screen-time rules and wasted counselling time; a Type II error would leave struggling heavy users unflagged. Neither cost dominates, so we kept the conventional level.

## 2. Data and preprocessing

Source: Shamim (2025), *Social Media Addiction vs Relationships* (Kaggle; file Students Social Media Addiction.csv), 705 rows, 13 self-reported variables, students aged 18–24, no missing values. Recruitment method, sampling frame and consent process are not documented (treated as a convenience sample); the file is de-identified. Student IDs 242 and 274 are identical in every answer; one was removed (**n = 704**). 110 countries: 79 (72%) contribute a single student, and the 31 countries with two or more students hold 625 students. For DML, Country was recoded to the 10 largest countries plus "Other".

| Variable | Type / range | Role | Reason |
|---|---|---|---|
| Avg_Daily_Usage_Hours | Continuous, 1.5–8.5 h | Treatment (X) | The conventional wisdom is about time spent |
| Mental_Health_Score | Integer, observed 4–9 (1–10 scale) | Outcome (Y) | Replaces GPA, which is not in the data |
| Age, Gender, Academic_Level, Relationship_Status | Background | Control | Fixed before usage; can confound, not caused by usage |
| Country | 110 categories | Control / cluster | Strongest confounder; breaks independence |
| Most_Used_Platform | 12 categories | Moderator | Imperfect proxy for purpose of use, not a direct measure |
| Sleep_Hours_Per_Night | Continuous | Mediator | Usage can reduce sleep; not controlled for the total effect |
| Addicted_Score, Conflicts_Over_Social_Media | Scores / counts | Not controlled | Likely consequences of usage (bad controls) |
| Affects_Academic_Performance | Yes / No | Secondary outcome | Closest proxy for the original GPA question |

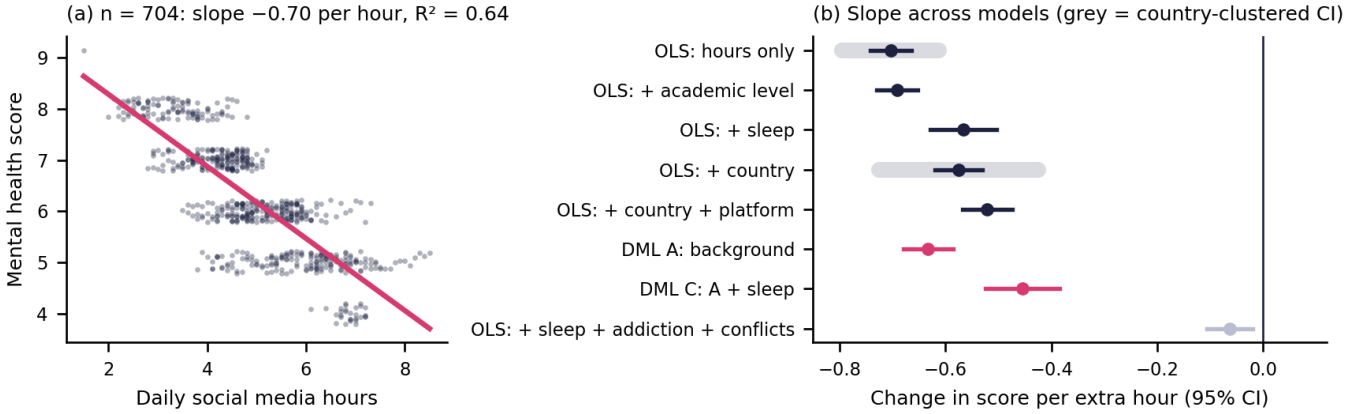
## 3. Methods

| Framework level | Method | Implementation details |
|---|---|---|
| Sampling distribution / Inference | OLS, two-sided t-test, 95% CI | df = 702, t* = 1.963. Country breaks independence, so CR1 cluster-robust SEs (110 clusters) are reported for the pooled slope and for the within-country (country fixed effects) slope. Shapiro–Wilk on residuals; Spearman rank correlation as an ordinal check. |
| Prediction | Held-out validation | 200 random 80/20 splits (seeds 0–199) and 200 repeats of 5-fold splits that hold out whole countries; test $R^2$ and RMSE vs a train-mean baseline. Prediction intervals: $\hat{y} \pm t^* \cdot s\sqrt{(1 + 1/n + (x-\bar{x})^2/S_{xx})}$. |
| Inference (heterogeneity) | Per-platform OLS; nested F-test | Bootstrap CIs (2,000 resamples). F-test: platform-specific vs common slope with platform intercepts (n ≥ 10, varying outcome). Same-country version: x and y demeaned by country within each platform, compared across the platforms where students share countries (WhatsApp and LinkedIn are not identified). Bonferroni across the 9 platform slope tests (α = 0.0056). |
| Reasoning | Fixed effects; mediation; sensitivity | 110 country dummies. Sleep mediation by product of coefficients a·b, 1,000 bootstrap resamples. Robustness value for an unmeasured confounder (Cinelli & Hazlett, 2020) from the simple-regression t-statistic. |
| Reasoning | Double/debiased ML (partially linear model) | $Y = \theta X + g(W) + \varepsilon$, $X = m(W) + v$ (Chernozhukov et al., 2018). Random forests (100 trees, min leaf 5; 8 in subgroups), 5-fold cross-fitting, 3 repetitions, median aggregation; influence-function SE that assumes independent students. |

## 4. Results

| Model | Estimate per hour | 95% CI | $R^2$ |
|---|---|---|---|
| OLS: hours only | −0.704 | −0.74 to −0.66 (clustered: −0.80 to −0.61) | 0.642 |
| OLS: + academic level | −0.691 | −0.73 to −0.65 | 0.657 |
| OLS: + sleep (direct effect) | −0.566 | −0.63 to −0.50 | 0.657 |
| OLS: + country fixed effects (within-country) | −0.575 | −0.62 to −0.53 (clustered: −0.73 to −0.42) | 0.848 |
| OLS: + country + platform | −0.521 | −0.57 to −0.47 | 0.874 |
| OLS: + sleep + addiction + conflicts | −0.063 | −0.11 to −0.02 | 0.896 |

| Model | Estimate per hour | 95% CI | $R^2$ |
|---|---|---|---|
| DML A: background (age, gender, level, relationship, country) | −0.633 | −0.68 to −0.59 | – |
| DML B: A + platform | −0.589 | −0.65 to −0.53 | – |
| DML C: A + sleep | −0.454 | −0.52 to −0.38 | – |
| DML D: A + addiction + conflicts | 0.056 | 0.01 to 0.10 | – |




Validation: random splits give mean test $R^2$ 0.635 (range 0.484–0.730), RMSE 0.662 vs baseline 1.104; holding out whole countries gives 0.623 (0.588–0.638), RMSE 0.678. Adjusted $R^2$: hours 0.641, country 0.629, both 0.819. Platform heterogeneity: $F(8, 650) = 3.59$, $p = 0.0004$ for all students; among students from the same country $F(3, 432) = 13.0$, $p < 0.0001$ (Facebook −0.74, Twitter −0.91, Instagram −0.36, TikTok −0.31); WhatsApp is 0 by construction (scores are constant within its four countries) and LinkedIn is not identified. All raw platform slopes pass Bonferroni (max $p = 0.0014$), but WhatsApp's and LinkedIn's reflect between-country differences; academic level $p = 0.81$; gender $p = 0.31$. Mediation: $a = −0.709$, $b = 0.194$, indirect = −0.138 (95% CI −0.177 to −0.104), 19.6% of total. Spearman $\rho = −0.802$ (Pearson −0.801). Sensitivity: robustness value 0.71 to erase the slope, 0.48 to halve it. DML subgroups: Undergraduate −0.63, Graduate −0.68, High School −0.70, Female −0.63, Male −0.69, Instagram −0.66, TikTok −0.54, Facebook −0.66; WhatsApp could not be estimated because scores are constant within its countries. Secondary outcome: students reporting academic harm use 5.54 vs 3.80 h (Welch $t = 25.4$, $p < 0.001$).

*Model D: controlling for addiction and conflicts removes the effect because both are probably downstream of usage and strongly tied to the outcome ($r = −0.95$ for addiction). We report it only to show why such variables must not be controlled.*

## 5. Limitations and threats to validity

- **External validity:** undocumented, non-random sample; 72% of countries contribute one student, so the within-country estimate rests on 31 countries. Held-out validation stays inside this dataset.
- **Internal validity:** cross-sectional self-reports; reverse causation and unmeasured confounding (stress, workload, purpose of use) remain. DML adjusts only for measured covariates, its intervals ignore clustering, and no valid instrument was available.
- **Measurement:** platform is an imperfect proxy for purpose of use, so platform differences do not show that passive and active use differ.
- **Data quality:** identical scores for all users within each of LINE, KakaoTalk and VKontakte and within 13% of countries with $n \geq 2$; academic-impact answers move from 0% to 100% across usage bands; correlations far stronger than the association found in a study of 355,358 adolescents (Orben & Przybylski, 2019). A public dataset with the same variable names is documented as statistically generated from an original survey (Badawy, n.d.); we found no such documentation for our file.
- **Ethics:** public, de-identified data with an undocumented consent process; results should not inform decisions about individual students.

## 6. Reproducibility

Repository: **analysis.py** (all computations, writes results.json), **results.json**, **data/**, **index.html** (website), **tools/** (page and document builders). Python 3 with pandas, numpy, scipy, scikit-learn, matplotlib, reportlab. Run **python analysis.py**, then **python tools/build_page.py**, **python tools/build_docs.py** and **python tools/build_maps_report.py**. All random procedures use fixed seeds; re-running analysis.py from a clean copy reproduced results.json exactly.

## References


*Shamim, A. (2025). Social Media Addiction vs Relationships [Data set]. Kaggle. https://www.kaggle.com/datasets/adilshamim8/social-media-addiction-vs-relationships*

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*Orben, A., & Przybylski, A. K. (2019). The association between adolescent well-being and digital technology use. Nature Human Behaviour, 3(2), 173–182. https://doi.org/10.1038/s41562-018-0506-1*